

Integer multiplication, just for fun

An experiment with AI-generated math

An experiment by `odepoint`*

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Abstract

These notes record a just-for-fun attempt to see how far AI can push an integer-multiplication bound. They document a hobby project, not a research paper. The generated argument is largely unverified: the accompanying scripts check finite identities and arithmetic, but do not validate the full mathematical argument. The work builds on Douglas Colkitt’s repository and OpenAI’s paper, cited below.

The proposed change replaces the side correction circuit in the complex finite network by reusable sums of triple indicators sharing two fixed points. Within each batch these indicators are orthonormal over $\mathbb{F}_2$. A bounded batch size provides explicit norm-one witnesses for every new complementary residual, in both circuit orientations. Together with a redundant-center elimination and first/third-stage auxiliary reuse, the construction at $h = 24$ has 4,496,369,044,992 roles and supports the complex saving $a_c = 4 \cdot 10^{-9}$. The previously constructed bit network is unchanged. Under the retained compact-control and multiplication interfaces, exact rational parameters give

$$T(n) = O(n(\log n)^{1-\kappa}), \quad \kappa = \frac{5929220328}{10^{19}} = 5.929220328 \cdot 10^{-10} > 2^{-31}.$$

This is about 7.14364 times the saving at the repository revision examined. These are conditional asymptotic exponent comparisons, not measured runtime improvements. The proof and exact certificates do not independently establish the complete upstream theorem.

1 Baseline, dependencies, and the change

The baseline is Douglas Colkitt’s *integer-mult-bounds* [1], commit

6e564879f51ae16f23d392e9e196c605f36d90df.

Its strongest stated witness is $83/10^{12} = 8.3 \cdot 10^{-11} > 2^{-34}$. This is newer than the earlier 2^{-59} witness. The baseline retains OpenAI’s *Integer multiplication below $n \log n$* [2], pinned at

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

The model is a fixed finite alphabet on a fixed finite number of one-dimensional Turing tapes. Source attribution, licenses, and the pinned manuscript are preserved.

The present result assumes the pinned manuscript’s stream operations, bit-transfer and complex-phase transfer, analytic estimates, prime selection, Gaussian resampling, synthetic transforms, and final rounding. It also retains the repository’s paired $h = 50$ bit circuit, nonadjacent routing, compact dirty-control movement, complete temporary reservations, local exceptional repair, generalized stopping

*Put together by Owen DePoint using AI-generated material from OpenAI models. This project is a fork of Douglas Colkitt’s *integer-mult-bounds*; the underlying paper is by OpenAI. This experiment has no independent review or endorsement from their authors.

guard, and tightened Gaussian choice. The latter written arguments have the same conditional status here as in the baseline. The new mathematical work concerns the finite complex circuit, its two binary frame assignments, and the parameter comparisons resulting from that circuit.

The old complex motif at $h = 25$ certifies only $a_c = 4.18 \cdot 10^{-10}$, making its stopped leaves the limiting term. The new motif certifies $a_c = 4 \cdot 10^{-9}$, larger than the retained bit saving. This changes which term controls the completed layer.

Quantity	Baseline complex motif	New complex motif
Ground size h	25	24
Ambient dimension m	15,625	13,824
Central roles per invocation	26	24
Side roles per invocation	3,693,800	546,748
Total roles W	58,645,352,620,000	4,496,369,044,992
Certified complex saving	$4.18 \cdot 10^{-10}$	$4 \cdot 10^{-9}$

The side-role comparison changes h as well as the circuit. At a common $h = 24$, the original side count is 2,819,432, so the new sum graph alone uses about 5.157 times fewer side roles. Total roles also benefit from using the same auxiliary bank at stages one and three.

2 The new finite complex circuit

2.1 A complex network from orthogonal pair-star sums

The following construction changes the complex scalar network and its binary labels. The bit network used for address movement is unchanged. Let $h > 6$ be even, use ground coordinates $\{1, \dots, h\}$, let $\mathcal{T} = \binom{\{1, \dots, h\}}{3}$, and put

$$v = \binom{h}{3}, \quad N = v^3, \quad m = h^3, \quad F = \mathbb{F}_2^h, \quad \mathcal{F} = F^{\otimes 3}.$$

The form on F is the ordinary binary dot product. For a triple T , write t_T for its indicator. Thus t_T has binary norm one, and $t_S \cdot t_T = 0$ whenever $|S \cap T|$ is even. All scalar arithmetic below takes place in $\mathbb{Z}[i, 1/2]$; the binary field is used only for labels and address frames. The two data banks X_a, Y_a have indices $a = (a_1, a_2, a_3) \in \mathcal{T}^3$. Write $u_a = t_{a_1} \otimes t_{a_2} \otimes t_{a_3}$ and $U_a = \langle u_a \rangle$. The choice of one-based ground coordinates is only a relabeling of the original interface.

Partitioning the side corrections. Give the ground set its natural order. Write each source triple uniquely as $T = \{a, b, k\}$ with $a < b < k$. For each fixed pair (a, b) , partition the ordered list of possible $k > b$ into consecutive batches, each of size at most $h - 6$. These batches partition the source triples. Fix one batch K and a target triple S , and put $r = |S \cap \{a, b\}|$. The contribution of this batch to the side injection is

$$\begin{cases} \frac{1}{2} \sum_{k \in K \setminus S} x_{\{a, b, k\}}, & r = 0, \\ -\frac{1}{2} \sum_{k \in K \cap S} x_{\{a, b, k\}}, & r = 1, \\ -\frac{1}{2} \sum_{k \in K \setminus S} x_{\{a, b, k\}}, & r = 2. \end{cases} \quad (\text{PS1})$$

An empty sum is omitted. For $r = 0$, only intersection-zero sources survive; for $r = 1$, only intersection-two sources survive; for $r = 2$, the intersection-two sources survive and the possible diagonal source $T = S$ is excluded. Summing (PS1) over the disjoint source batches therefore gives exactly

$$- \sum_{\substack{T \neq S \\ |S \cap T| \text{ even}}} \frac{|S \cap T| - 1}{2} x_T. \quad (\text{PS2})$$

The signs in (PS1) belong to the output injections. All intermediate nodes compute sums with coefficient +1.

A specified cancellation-free sum graph. For each batch, represent a nonempty source subset by a bit mask M , with the bits ordered by increasing k . A singleton mask is an input node. For a mask with at least two set bits, let ℓ be the lowest set-bit index and let u be one more than the highest set-bit index. Set $j = \lfloor (\ell + u)/2 \rfloor$ and split M into its bits below j and its bits at least j . Both pieces are nonempty and disjoint. Compute their sums recursively and add the results. Intern every mask within this batch, so a previously computed formal sum is reused. Process batches in the order of their anchor pair and then their first k , and process target triples in lexicographic order. Request precisely the nonempty supports in (PS1), keeping one output use per target and batch, even when two uses request the same sum. This completely specifies the finite graph; its children are defined using the occupied-mask endpoints, not the endpoints of the entire batch.

Every addition has disjoint input supports. Consequently a source has coefficient one in a node exactly when it belongs to that node's mask; there is no cancellation and no repeated contribution. Let c_{add} be the number of distinct addition nodes and let q_{out} count output uses, including multiplicities. The same graph is used for every invocation of a stage. Its input triples are partitioned between batches, so no node combines different batches.

Reversible roles and signed restoration. Treat each use by an addition and each designated output use as an outgoing edge. At an input node allocate one role for every outgoing edge, select one as a pivot, and add the pivot into the others. At an addition node reuse its first incoming role as the pivot for one outgoing edge; allocate a new role for every other outgoing edge. Add the second incoming value to the pivot and then add the pivot into the new outgoing roles. The second incoming role is thereafter inactive during the forward graph. Every node operation is invertible by reversing its additions. Roles are identified only along directed paths, so two simultaneously used incoming roles remain distinct.

There are $2c_{\text{add}} + q_{\text{out}}$ outgoing uses and precisely one role identification per addition. Thus the side computation uses

$$R_{\text{side}} = c_{\text{add}} + q_{\text{out}} \quad (\text{PS3})$$

roles. Write $\mathcal{L}$ for the resulting invertible mixer on these roles, $\mathcal{V}$ for copying the physical sources into the input pivots, and $\mathcal{J}$ for injecting the designated output slots into their targets with the coefficients in (PS1). On a zero side vector, $\mathcal{J}\mathcal{L}\mathcal{V}$ is (PS2).

Let $\mathcal{G}, \mathcal{R}$ denote the central gather and scatter defined below. An operation written $+\mathcal{J}$ or $-\mathcal{J}$ means adding or subtracting the indicated injection into the targets, leaving all its controls unchanged; the same convention applies to $\mathcal{V}, \mathcal{G}$ and $\mathcal{R}$. The forward chronological schedule is

$$\mathcal{L}, -\mathcal{J}, \mathcal{L}^{-1}, -\mathcal{R}, +\mathcal{V}, +\mathcal{G}, +\mathcal{R}, \mathcal{L}, +\mathcal{J}, \mathcal{L}^{-1}, -\mathcal{G}, -\mathcal{V}. \quad (\text{PS4})$$

For arbitrary initial side vector z and central vector c_0 , the net target change is

$$-\mathcal{J}\mathcal{L}z - \mathcal{R}c_0 + \mathcal{R}(c_0 + \mathcal{G}x) + \mathcal{J}\mathcal{L}(z + \mathcal{V}x) = (\mathcal{J}\mathcal{L}\mathcal{V} + \mathcal{R}\mathcal{G})x.$$

The final two operations restore the centers and the side vector exactly. The minus signs in (PS4) are essential over the complex scalar ring.

Only h central coordinates are needed. Use center roles $C_1, \dots, C_{h-1}, C_*$. Gather by adding

$$(\mathcal{G}x)_i = \sum_{T \ni i} x_T \quad (i < h), \quad (\mathcal{G}x)_* = \sum_T x_T.$$

Define the scatter at target S by

$$(\mathcal{R}c)_S = \begin{cases} \frac{1}{2} \left(\sum_{i \in S} c_i - c_* \right), & h \notin S, \\ c_* - \frac{1}{2} \sum_{\substack{i < h \\ i \notin S}} c_i, & h \in S. \end{cases} \quad (\text{PS5})$$

The omitted gather coordinate equals $3(\mathcal{G}x)_* - \sum_{i < h} (\mathcal{G}x)_i$. Substitution into the original scatter proves

$$(\mathcal{R}\mathcal{G})_{S,T} = \frac{|S \cap T| - 1}{2}.$$

This identity is asserted for gather increments, not for arbitrary initial center values. The latter cancel in (PS4), so no initial consistency condition is imposed on the centers. Combining this identity with (PS2) gives $\mathcal{J}\mathcal{L}\mathcal{V} + \mathcal{R}\mathcal{G} = I$. Thus every forward invocation is $y \leftarrow y + x$ and restores arbitrary scratch. The inverse invocation uses the reversed inverse schedule. The three stage orientations remain forward $X \rightarrow Y$, inverse $Y \rightarrow X$, and forward $X \rightarrow Y$, and hence send (X, Y) to $(-Y, X)$.

Common frames in the two orientations. Use the original stage decomposition

$$A = F^{\otimes(j-1)} = B \perp P, \quad P = \langle t_{a_1} \otimes \dots \otimes t_{a_{j-1}} \rangle, \quad Q = \langle t_{a_{j+1}} \otimes \dots \otimes t_{a_3} \rangle.$$

Empty tensor products are the one-dimensional field. Both P and Q have norm-one generators. Suppress Q temporarily and write

$$D_0 = B \otimes F, \quad D_1 = A \otimes F, \quad D_U = (B \otimes F) \perp (P \otimes U), \quad X_t = D_{\langle t \rangle}, \quad Y_t = D_{t^\perp}.$$

All these labels are tensored with Q in $\mathcal{F}$.

At a graph node z , let U_z be the span of the source indicators in its mask. Two distinct triples in its batch share exactly the two anchors, so their binary indicators are orthogonal. Each has norm one. Thus these indicators form an orthonormal basis of U_z . Every U_z is nondegenerate, and along an edge $z \rightarrow w$ one has $U_z \subset U_w$. At a designated output for target S , all contributing source indicators are orthogonal to t_S , by (PS1); hence $U_z \subset t_S^\perp$.

The common frames for the twelve forward operations in (PS4) are, in order,

$$D_0, D_0, D_0, D_0, X_{t_T}, D_1, D_0, D_{U_z}, Y_{t_S}, D_1, D_1, D_1. \quad (\text{PS6})$$

The X copy gates and Y injection gates are grouped by their physical data triple; the middle mixer is grouped by graph node. At each such gate the same displayed label is assigned to every incident role. All early mixer gates use D_0 and all late cleanup gates use D_1 . An input or output node that otherwise gives an identity operation may be retained as an identity gate to specify these incidences.

In the reversed invocation the middle computation is $\mathcal{L}^{-1}$. Assign its node z the complement frame $D_{U_z^\perp}$. Reversing an edge gives $U_w^\perp \subset U_z^\perp$. At an original output $\langle t_S \rangle \subset U_z^\perp$, and at an original

input $U_z^\perp = t_T^\perp$. These are precisely the reversed physical X and Y attachments. In chronological order the reversed operations and labels are

step	operation	label	step	operation	label
1	$+\mathcal{V}$	D_0	7	$-\mathcal{G}$	D_0
2	$+\mathcal{G}$	D_0	8	$-\mathcal{V}$	Y_{t_T}
3	$\mathcal{L}$	D_0	9	$+\mathcal{R}$	D_1
4	$-\mathcal{J}$	X_{t_S}	10	$\mathcal{L}$	D_1
5	$\mathcal{L}^{-1}$	$D_{U_z^\perp}$	11	$+\mathcal{J}$	D_1
6	$-\mathcal{R}$	D_1	12	$\mathcal{L}^{-1}$	D_1

(PS7)

The logical source in (PS7) is the physical Y bank. Pointwise scalar coefficients, including the signs and halves, commute with the common array frame exactly as in the original frame identity.

Binary residuals, including entering and retired roles. The following argument checks the additional condition needed by the complex phase transfer: every nonzero residual must have an orthonormal basis, not merely be nondegenerate. The source support of a node occupies at most $(h-6)+2 = h-4$ ground coordinates. Adjoining one target triple occupies at most $h-1$ coordinates. There is therefore a coordinate unit outside either union. Such a unit belongs to $U_z^\perp$, and in the second case it also belongs to $t_S^\perp$; its norm is one.

All the following residuals are nondegenerate. For nested nondegenerate spaces this follows by orthogonally splitting the larger space into the smaller one and its residual. Suppressing the common factor P , the new forward residual classes are

transition	residual	norm-one witness if nonzero
$D_0 \rightarrow D_U$	U	a source indicator
$D_U \rightarrow D_V$ ($U \subset V$)	$V \cap U^\perp$	an indicator in the support difference
$D_U \rightarrow Y_{t_S}$	$t_S^\perp \cap U^\perp$	a unit outside both supports
$D_U \rightarrow D_1$	$U^\perp$	a unit outside the source support.

(PS8)

The second row includes an input line entering a later sum. It has the indicators in V 's support but not U 's support as an orthonormal basis. The first row also covers a newly allocated outgoing role; the last row covers a second input retired after an addition. A direct $D_0 \rightarrow D_1$ transition has residual F .

For the reversed computation the corresponding complete list is

transition	residual	norm-one witness if nonzero
$D_0 \rightarrow D_{U^\perp}$	$U^\perp$	a unit outside the source support
$X_{t_S} \rightarrow D_{U^\perp}$	$t_S^\perp \cap U^\perp$	a unit outside both supports
$D_{V^\perp} \rightarrow D_{U^\perp}$ ($U \subset V$)	$V \cap U^\perp$	an indicator in the support difference
$D_{U^\perp} \rightarrow D_1$	U	a source indicator.

(PS9)

Roles retired in the forward mixer become entering roles in its inverse, and newly allocated forward roles become retired inverse roles. Thus the first and last rows of (PS9) are necessary; the two orientations do not assume zero values on such roles. At an original input the final attachment to Y_{t_T} is an equality of labels. Tensoring a displayed witness with the norm-one generators of P and Q gives a norm-one witness for the actual residual in $\mathcal{F}$.

The remaining data and central residuals are those of the original tensor stage table: zero spaces, copies of $P \otimes F$, copies of $P \otimes t^\perp$, and copies of $B \otimes t^\perp$, all tensored with Q . Initial auxiliary edges have residual $B \otimes F \otimes Q$; final auxiliary edges have residual $(A \otimes F) \otimes Q^\perp$. When B or $Q^\perp$ is nonzero

it contains a coordinate unit outside the support of its defining earlier or future tensor indicator. Those supports have size $3^j < h^j$. A triple complement contains a coordinate unit outside the triple. These observations give norm-one witnesses for every nonzero remaining tensor residual. In particular all data boundaries are the original ones. The sole decreases are the h central-role returns from D_1 to D_0 in each invocation, each of dimension h .

A nondegenerate symmetric binary space containing a norm-one vector has an orthonormal basis, by the elementary binary-space argument in the original residual lemma. Equations (PS8)–(PS9) and the preceding tensor witnesses therefore verify the full orthonormal-residual hypothesis of the complex phase interface. No positivity property of the rational bit labels is being substituted for this binary condition.

Reusing the first-stage auxiliary bank in stage three. Pair the h ground coordinates consecutively and let π send a triple to its image under the permutation swapping the two members of every pair. The intersection of a triple and its image has size zero or two, so $t_A \cdot t_{\pi(A)} = 0$. The map π is a bijection of triples. Identify the entire auxiliary bank, including its h centers, of the stage-1 invocation with fixed triples (A, B) with the auxiliary bank of the stage-3 invocation with fixed triples $(B, \pi(A))$, retaining each local role index. This is a bijection of invocation banks. The two invocations occur at different stages and restore arbitrary scratch, so no simultaneous values are identified.

Every auxiliary role now first has label $D_0 \otimes Q$ and last has label $D_1 \otimes Q$. The new interstage join is

$$E = F \otimes \langle t_A \rangle \otimes \langle t_B \rangle \subset H = \langle t_B \otimes t_{\pi(A)} \rangle_{F \otimes F}^\perp \otimes F. \quad (\text{PS10})$$

Both spaces are nondegenerate, and the inclusion follows from $t_A \perp t_{\pi(A)}$. The residual $H \cap E^\perp$ is nondegenerate. Choose a ground coordinate j outside $A \cup \pi(A)$, and any coordinates r, k . The norm-one tensor $e_r \otimes e_j \otimes e_k$ belongs to both H and $E^\perp$. Thus the new join residual also has an orthonormal basis. It introduces no decreasing edge. It replaces the old edges $E \rightarrow \mathcal{F}$ and $0 \rightarrow H$ and reduces their total absolute dimension change by exactly m per removed role.

The coefficient-depth allowance still applies. Every elementary arithmetic operation uses only coefficients $0, \pm 1, \pm 1/2$. One application of $\mathcal{L}$ or its inverse contains at most $2R_{\text{side}}$ scalar additions or subtractions: the outgoing-use count is $2c_{\text{add}} + q_{\text{out}}$, and the input fanouts save one operation per input. There are four mixer applications in (PS4). The two signed output injections cost at most $4q_{\text{out}}$ additions and halvings; the two source copies cost at most $2v$ additions; the two gathers cost at most $8v$ additions; and the two scatters cost at most $4vh$ additions and halvings. Thus an invocation uses at most

$$12R_{\text{side}} + 4vh + 10v$$

elementary coefficient operations. Reversing the schedule has the same bound. A conservative allowance for the complete three-stage scalar circuit and its scalar endpoint corrections is therefore

$$D_{\text{scalar}} = 3v^2(12R_{\text{side}} + 4vh + 10v) + 8W. \quad (\text{PS11a})$$

This counts all operations in the finite scalar circuit, so in particular it bounds every coefficient dependency path there. For each inverse recursive child, the two $Z^{\otimes f}$ corrections are each one sign selected by address parity, and $(-i)^f$ costs at most a sign and a multiplication by i . The retained bound of four coefficient operations per child therefore applies. Its computation of address parities is descriptor work. The complete nonrecursive node allowance is

$$D_{\text{node}} = D_{\text{scalar}} + 4s < 64(W + m + 1)^3, \quad (\text{PS11b})$$

where the strict inequality follows by integer comparison at the values below. Source, sink and bank corrections were already included in the $8W$ term; the original bound for them is $4W + 4 \leq 8W$.

For completeness, the number of edges also stays explicitly bounded. Refine each grouped scalar gate into two-role shears, retaining its common label throughout that refinement. Its number of touched-role incidences is at most twice the operation count used above. An identity incidence may be retained for each side role in each of the four mixer applications. A physical role participates in at most two shared invocations, so there are at most $8W$ such additional incidences. Each role contributes one more edge than its number of gate incidences. Hence the total edge count is at most

$$Q_{\text{edge}} = 2D_{\text{scalar}} + 9W < 100W. \quad (\text{PS11c})$$

All new edges created by a refinement have zero residual. For every nonzero residual edge its two binary changes of basis need at most $2(m(m-1) + 3m)$ binary row additions. These operations permute complete coefficient encodings and add zero coefficient depth. They are charged to the compact-control movement bound in time, not to magnitude or denominator growth. Thus they require no additional term in (PS11b).

Consequently the retained guard recurrence may use its existing node charge $E = 64(W + m + 1)^3$, now evaluated with the new network constants. The recursive residual calls are charged separately by $sA(e/m)$, as before. Address permutations and compact-control repair introduce no additional coefficient arithmetic.

Finite counts and the phase interface. There are $3v^2$ invocations. First/third-stage reuse leaves two complete auxiliary banks per pair of fixed triples, giving

$$W = 2N + 2v^2(R_{\text{side}} + h), \quad L_{\text{dec}} = 3v^2h^2, \quad s = Wm - 2N + 2L_{\text{dec}}. \quad (\text{PS11})$$

Here L_{dec} still counts central decreases in all three stages; sharing a central role does not remove either of its two separate returns. The signed sum of edge dimension changes telescopes to $Wm - 2N$, and every decrease has just been listed, proving the formula for s .

At $h = 24$, exhaustive exact support generation by the stated mask rule gives

$$\begin{aligned} v &= 2024, & m &= 13824, & N &= 8291469824, \\ c_{\text{add}} &= 77454, & q_{\text{out}} &= 469294, & R_{\text{side}} &= 546748, \\ W &= 4496369044992, & L_{\text{dec}} &= 7078883328, \\ s &= 62157803252796416, & Wm - s &= 2425172992 > 0. \end{aligned} \quad (\text{PS12})$$

The companion checker verifies every mask split, every designated signed output, the complete side matrix, and these integer counts. These finite checks support the explicit graph and its arithmetic; the general frame and tape claims use the arguments given here and the retained interfaces.

Proposition 1. *The $h = 24$ construction has W complex roles and total binary phase-residual length s given in (PS12). It performs the original signed bank exchange, restores arbitrary auxiliary values, and satisfies the original complex phase interface on any positive number of columns. In particular*

$$\frac{s}{W} < m^\sigma, \quad \sigma = 1 - \frac{4}{10^9}.$$

Proof. The signed scalar schedule was proved in (PS4)–(PS5). Every gate has a common nondegenerate frame, every edge joins comparable labels, and every nonzero residual has an orthonormal basis by (PS8)–(PS10). The original binary phase factorization therefore contributes exactly one forward or inverse translation kernel per residual basis vector, totaling s . The input/output data labels remain

$U_a, 0$ and $\mathcal{F}, U_a^\perp$ for X_a, Y_a , respectively. Every surviving auxiliary role starts at zero, ends at $\mathcal{F}$, and is fixed by the scalar routing. The original signed endpoint correction consequently applies on every role. More explicitly, u_a has norm one and weight 27; the $X_a \rightarrow Y_a$ route has the same 27 inverse factors per column. The original source and sink Z corrections, including the scalar i^{27k} on k columns, turn these into forward factors. Negating the physical X outputs and undoing the bank permutation then gives the required $C^{\otimes mk}$ on every role. Finally the exact arithmetic comparisons

$$\sum_{j=0}^{15} \frac{(48/5)^j}{j!} > 13824 = m, \quad \eta := \frac{Wm - s}{Wm} = \frac{37}{948319488} > \frac{4}{10^9} \frac{48}{5}$$

give $\log m < 48/5$ and $\eta > (4/10^9) \log m$. The strict inequality $e^{-x} > 1 - x$ for $x > 0$ now gives $m^{1-4/10^9} > m(1 - \eta) = s/W$. $\square$

3 Exact recurrence exponents

The complex counts above give

$$\eta_c = 1 - \frac{s}{Wm} = \frac{37}{948319488}.$$

The supplied exact logarithm enclosure proves

$$\log m < \frac{9535}{1000}, \quad \eta_c > \frac{4}{10^9} \frac{9535}{1000}.$$

For $x > 0$, $e^{-x} > 1 - x$. Hence

$$m^{1-a_c} = me^{-a_c \log m} > m(1 - a_c \log m) > m(1 - \eta_c) = \frac{s}{W}, \quad a_c = \frac{4}{10^9}.$$

This establishes the required strict complex recurrence exponent $\sigma = 1 - a_c$. An exact enclosure of the actual recurrence saving is approximately $4.0922725765 \cdot 10^{-9}$, but the simpler certified $4 \cdot 10^{-9}$ already suffices.

The bit graph remains the existing paired construction at $h_b = 50$, with $m_b = 125000$ and exact relative deficit

$$\eta_b = \frac{23}{661055000}.$$

For the readable witness below one may retain $a_b = 296/10^{11}$. For the slightly stronger final witness, certify

$$a_b = \frac{29646101715649}{10^{22}} = 2.9646101715649 \cdot 10^{-9}.$$

If L_b^+ is the supplied rational upper bound on $\log(125000)$, exact arithmetic establishes

$$\eta_b + \frac{\eta_b^2}{2} > a_b L_b^+.$$

The positive Taylor tail gives $-\log(1 - \eta_b) > \eta_b + \eta_b^2/2$. It follows that the same bit circuit supports $\tau = 1 - a_b$. This sharpening changes no gate, role, frame, or tape algorithm. Using just the linear Taylor term is insufficient for this sharper numerical value; the verifier specifically checks that distinction.

4 Completed layers and two explicit parameter witnesses

The retained compact-control recurrence gives the internal, leaf, and reservation exponents

$$\chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\}, \quad \chi_{\text{leaf}} = \sigma + \beta(1 - \sigma), \quad \chi_{\text{reserve}} = \max\{1 - c, 0\}.$$

Here $a_c > a_b$, so $\sigma < \tau$ and $\chi = \tau$. Retaining the old formula without the maximum would incorrectly overstate the gain. We require

$$\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1, \quad \max\{\chi_{\text{leaf}}, \chi_{\text{reserve}}\} < \lambda'.$$

For rational $0 < \beta < 1$ and $\zeta > 0$ the guard has $C_1 = 5 - 4\beta + \zeta$. Its node constant is evaluated using the *new* m, W, s , not the baseline complex counts. The finite circuit bound in the construction and $2 \leq s < m^5$ supply its hypotheses.

We use $c = 1$, so $K = d$. The compact front and back fields each reserve $O(\log p)$ coordinate chunks, and the row reservations use $O(\log d)$ chunks. Their total remains $o(d)$. Reservation preprocessing has exponent zero, while $\epsilon(1 + c) = 2\epsilon < 1$ ensures $K = o(\ell)$ with $\ell = \Theta(p^{1-\epsilon})$. Thus $c = 1$ is within the retained layout proof; no free temporary address coordinates or new tape operation are being assumed.

4.1 A readable witness

Using $a_b = 296/10^{11}$, choose

$$\begin{aligned} \epsilon &= \frac{19999}{100000}, \quad c = 1, \quad \beta = \frac{1}{4}, \quad \zeta = \frac{1}{100}, \quad C_1 = \frac{401}{100}, \quad \delta = \frac{1}{10^6}, \\ \lambda &= 1 - \frac{5919}{2 \cdot 10^{12}}, \quad \lambda' = 1 - \frac{2959}{10^{12}}, \quad \kappa_0 = \frac{59}{10^{11}} = 5.9 \cdot 10^{-10}. \end{aligned}$$

The leaf saving is $(1 - \beta)a_c = 3 \cdot 10^{-9}$, strictly larger than $1 - \lambda' = 2.959 \cdot 10^{-9}$. The internal saving is $a_b = 2.96 \cdot 10^{-9} > 1 - \lambda > 1 - \lambda'$. Also $\epsilon C_1 < 1$. The seven assembly margins are

$$\begin{aligned} g_1 &= 1 - \epsilon(1 + c), & g_2 &= \epsilon c a_b, \\ g_3 &= \epsilon(1 - \lambda'), & g_4 &= a_b(1 - \epsilon), \\ g_5 &= \frac{1}{4} - \delta - \frac{5}{4}\epsilon, & g_6 &= 1 - \delta - \epsilon, \quad g_7 = \epsilon. \end{aligned}$$

Exact arithmetic gives

$$\min_i g_i = g_3 = \frac{59177041}{10^{17}} = 5.9177041 \cdot 10^{-10} > \kappa_0.$$

All remaining setup and precision constraints are checked in the companion certificate. This already proves the conditional corollary $\kappa = 2^{-31}$.

4.2 The strongest supplied witness

Use the sharpened $a_b = 29646101715649/10^{22}$ and set $t = 10^{-12}$. Choose

$$\begin{aligned} \epsilon &= \frac{399999999}{2000000000}, \quad c = 1, \quad \beta = \frac{1}{10}, \quad \zeta = \frac{1}{100}, \quad C_1 = \frac{461}{100}, \quad \delta = 10^{-12}, \\ \lambda &= 1 - a_b(1 - t/2), \quad \lambda' = 1 - a_b(1 - t), \quad \kappa = \frac{5929220328}{10^{19}}. \end{aligned}$$

Now $(1 - \beta)a_c = 3.6 \cdot 10^{-9} > a_b$, so the leaves do not limit $1 - \lambda'$. The minimum margin is

$$G = \epsilon a_b(1 - t) = \frac{11858440656601639843694386501715649}{2 \cdot 10^{43}}.$$

In particular,

$$G - \kappa = \frac{601639843694386501715649}{2 \cdot 10^{43}} > 0, \quad g_5 = \frac{39}{62500000000} > G.$$

The other five margins are also strictly larger than κ . The guard exponent obeys

$$\epsilon C_1 = \frac{184399999539}{200000000000} < 1.$$

The unchanged Gaussian choice $\alpha = \lceil (32db)^{1/4} \rceil$ has exponent $(1 + \epsilon)/4$, and $\gamma = 2d\alpha^2$ has exponent $1/2 + 3\epsilon/2 < 1$. Prime selection retains $1 - 2\epsilon > 0$. The record suffix still grows faster than any fixed polynomial in p , and all rational exponent comparisons are fixed finite machine computations. The strict gap $G - \kappa$ absorbs the remaining fixed powers of $\log p$.

Theorem 1 (Conditional extension). *Under the retained upstream and compact-control interfaces identified above, the explicitly constructed finite network and parameter witness yield*

$$T(n) = O\left(n(\log n)^{1-5929220328/10^{19}}\right).$$

The saving lies strictly between 2^{-31} and 2^{-30} and exceeds the baseline $83/10^{12}$ by the factor $741152541/103750000 \approx 7.14364$.

5 A scoped ceiling and the next obstruction

This construction moves the bottleneck to the bit recurrence. Write $q = 1 - \lambda'$. The strict layer constraints imply $q < a_b$. For any feasible saving,

$$\kappa < \epsilon q, \quad \kappa < \frac{1}{4} - \delta - \frac{5}{4}\epsilon.$$

Eliminating ϵ yields

$$\kappa < \frac{q(1 - 4\delta)}{5 + 4q} < \frac{a_b}{5 + 4a_b}.$$

Conversely, with this complex saving, $\beta = 1/4$, $\zeta = 1/100$ and $c = 1$, take $q \uparrow a_b$, $\delta \downarrow 0$ and $\epsilon = (1 - 4\delta)/(5 + 4q)$; choose λ strictly between τ and λ' . The leaf and guard constraints stay slack and the other assembly margins do not bind. Thus $a_b/(5 + 4a_b)$ is the parameter supremum for this fixed certified bit saving and these retained constraints. It is about $5.9292203291 \cdot 10^{-10}$ for the sharpened a_b . Replacing a_b by the exact bit-graph saving and using rational logarithm enclosures changes only the last reported digits.

This ceiling is not a lower bound on integer multiplication, nor an optimality result for all finite networks. To cross the next dyadic milestone 2^{-30} within this approach, one needs a stronger bit primitive or a stronger downstream estimate, such as a changed Gaussian cost bound. Further reducing the complex circuit alone does not remove the present restriction.

6 Reproduction and verification boundary

The new implementation uses only Python’s standard library:

```
python3 scripts/complex_pair_star.py
python3 scripts/complex_pair_star_parameters.py
python3 scripts/make_complex_pair_star_patch.py
python3 -m unittest discover -s tests -p 'test_complex_pair_star*.py' -v
git apply --check --directory=upstream patches/complex-pair-star-31.patch
```

The main certificate records all 263 batches, all 77,454 additions, all 469,294 output uses, every one of the 4,096,576 ordered source–target coefficients, both compiled frame directions, residual unit witnesses, exact recurrence inequalities, and all seven assembly margins. The tests additionally exercise the complete signed invocation, its inverse, and a three-shear signed exchange with reused arbitrary dyadic scratch at $h = 8$. Real dyadic tests suffice for the complex scalar identities because all operations are linear with real dyadic coefficients and hence act independently on real and imaginary parts.

The new patch applies independently to the unmodified pinned manuscript. It incorporates the previously retained compact-control modifications and inserts this new complex interface and the sharpened bit comparison. The verifier does not implement the complete multiplication machine, and finite tests do not prove uniform Turing-tape estimates or the upstream analytic identities. Those remain the explicitly stated proof dependencies. No new unrestricted optimality, independent human review, or practical performance claim is made.

References

- [1] Douglas Colkitt. *A sharper exponent for integer multiplication*. CrocSwap/integer-mult-bounds, revision 6e564879f51ae16f23d392e9e196c605f36d90df, October 7, 2026. <https://github.com/CrocSwap/integer-mult-bounds>.
- [2] OpenAI. *Integer multiplication below $n \log n$* . Source pinned at revision adc7f1241b42e322a6451854ab7e4b4c146bf78a; see the repository’s upstream/manifest.json for source URLs and hashes. <https://github.com/openai/math>.